

How Did Shifts in Population Affect Home Prices During the Pandemic?

Federal Reserve Bank of Chicago — RA Candidate Data Task

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Summary of Findings

Metro areas that gained population during 2020–2021 experienced significantly stronger home price appreciation than those that lost residents. Across 380 matched Metropolitan Statistical Areas, the correlation between population change rate and home price appreciation is 0.576. In OLS regression, a one-unit increase in the population change rate (per 1,000 residents) is associated with approximately 0.36–0.37 percentage points of additional home price appreciation, significant at all conventional significance levels and stable across all model specifications. This is consistent with the “zoom town” narrative: remote work untethered workers from expensive urban areas, generating a demand shock in destination markets that housing supply could not quickly absorb.

1 Data

I Home Prices: Zillow ZHVI

I use the Zillow Home Value Index (ZHVI) at the Metropolitan Statistical Area (MSA) level, middle tier (33rd–67th percentile), covering all home types, seasonally adjusted and smoothed at monthly frequency. The dataset contains 894 MSAs after filtering to `RegionType == "msa"`. The middle tier is preferred over full-distribution indices because it captures the segment of the market most responsive to migration-driven household demand, avoiding distortion from luxury sales at the top or distressed properties at the bottom.

II Population Change: Census Bureau PEP

Population data was sourced from the Census Bureau Population Estimates Program (PEP), Vintage 2022, which provides annual MSA-level estimates with components of population change. After filtering to `LSAD == "Metropolitan Statistical Area"`, the file contains 384 MSAs. The key variable is `NPOPCHG`, total population change, equal to natural increase (births minus deaths) plus net migration (domestic and international combined).

`NPOPCHG` is a reasonable proxy for migration during this window for two reasons. First, natural increase was geographically uniform and COVID-suppressed during 2020–2021: elevated excess mortality and declining birth rates mean that cross-metro variance in `NPOPCHG` is driven overwhelmingly by migration rather than natural increase. Second, international migration was sharply curtailed by border restrictions in 2020–2021, making domestic migration the dominant non-natural component. With more time, the ideal variable would be net domestic migration specifically, isolated from ACS migration flow tables or IRS Statistics of Income county-to-county flow data.

III Merging

Zillow uses short metro names (e.g., “Houston, TX”) while the Census uses full CBSA names (e.g., “Houston-The Woodlands-Sugar Land, TX”). I matched entries on a key of primary city names *and* state abbreviations extracted from each label. An initial merge on city name alone produced false cross-state matches, most notably Austin, TX being matched to Austin, MN, which introduced noise and attenuated my regression estimates. Adding the state abbreviation as a second key eliminated these false matches. The improvement was material: the bivariate R^2 rose from 0.24 to 0.33 and the migration coefficient from 0.32 to 0.37. The final merged dataset contains 380 MSAs. The roughly 500 Zillow entries without a Census match are predominantly small micropolitan markets outside the MSA definition used by PEP, and not of use to this analysis.

2 Methodology

I Home Price Change

I computed the percentage change in ZHVI from a pre-pandemic baseline to a pandemic peak:

$$\Delta p_i = \frac{\bar{Z}_{i,\text{peak}} - \bar{Z}_{i,\text{pre}}}{\bar{Z}_{i,\text{pre}}} \times 100$$

where $\bar{Z}_{i,\text{pre}}$ is the 3-month average ZHVI for metro i over November 2019 – January 2020, and $\bar{Z}_{i,\text{peak}}$ is the 3-month average over October – December 2021. The three-month averages at each endpoint reduce the sensitivity to noise from a single-month choice. The window is chosen deliberately: January 2020 is the last pre-lockdown month; December 2021 captures the peak of the pandemic migration-driven demand shock before prices began to decelerate. Home price changes in the final sample range from -2.1% (Lake Charles, LA) to $+58.2\%$ (Boise City, ID), with a mean of 24.3% and median of 23.6% .

II Population Change Rate

I sum NPOPCHG over 2020 and 2021 and normalize by the April 1, 2020 Census base population:

$$\text{PopRate}_i = \frac{\text{NPOPCHG}_{i,2020} + \text{NPOPCHG}_{i,2021}}{\text{EstimatesBase}_{i,2020}} \times 1,000$$

Expressing the rate per 1,000 residents makes large and small metros directly comparable (e.g., Phoenix, AZ vs. Pocatello, ID). Population change rates in the sample range from $-54.9/1\text{k}$ (Lake Charles, LA) to $+62.1/1\text{k}$ (St. George, UT), with a mean of $4.8/1\text{k}$ and median of $3.3/1\text{k}$. Annual rates for 2020 and 2021 separately are also computed for robustness checks.

III Regression Specifications

I estimate three OLS cross-sectional specifications:

$$\Delta p_i = \alpha + \beta_1 \text{PopRate}_i + \varepsilon_i \tag{1}$$

$$\Delta p_i = \alpha + \beta_1 \text{PopRate}_i + \beta_2 \ln(Z_{i,\text{pre}}) + \varepsilon_i \tag{2}$$

$$\Delta p_i = \alpha + \beta_1 \text{PopRate}_{i,2021} + \beta_2 \ln(Z_{i,\text{pre}}) + \varepsilon_i \tag{3}$$

(2) is the preferred specification. The log of pre-pandemic home value controls for affordability constraints and unobserved amenity demand. (3) uses 2021-only population change as a robustness check, since 2021 carries a cleaner migration signal than 2020, when the initial lockdown shock briefly distorted flows before the broader suburban and Sun Belt trend emerged.

3 Results

I Regression Results

Table 1 reports estimates across all three specifications. The coefficient on population change rate is positive, statistically significant at $p < 0.01$ across all models, and stable at approximately 0.357–0.373. This stability is illustrated in Figure 4.

Table 1: OLS Regression Results: Population Change and Home Price Appreciation

	(1): Bivariate	(2): + Price Level	(3): 2021 Only
Pop. change rate (per 1,000)	0.371*** (0.027)	0.357*** (0.027)	—
Pop. change rate 2021 (per 1,000)	—	—	0.373*** (0.031)
ln(Pre-pandemic ZHVI)	—	3.770*** (0.836)	4.070*** (0.861)
Intercept	22.46 *** (0.389)	−23.73** (10.20)	−27.25** (10.52)
Observations	380	380	380
R^2	0.332	0.366	0.325

Standard errors in parentheses. ** $p < 0.01$; *** $p < 0.001$

Sources: Zillow ZHVI; Census Bureau PEP Vintage 2022.

The preferred specification (2) indicates that, holding pre-pandemic price level constant, each additional resident per 1,000 is associated with 0.36 percentage points of additional home price appreciation. An R^2 of 0.33–0.37 is strong for a single-variable cross-sectional regression, indicating population change alone explains roughly a third of the variation in home price appreciation across metros.

Notably, the coefficient on $\ln(\text{ZHVI}_{\text{pre}})$ is positive and significant across both specifications that include it. This means more expensive markets appreciated *more* in percentage terms even controlling for migration, contrary to simple mean reversion, and consistent with sustained demand for high-amenity superstar cities.

II Descriptive Results

Table 2 shows the five highest- and lowest-appreciating metros in the sample, illustrating the stark contrast between pandemic destinations and origin cities.

Table 2: Top 5 and Bottom 5 Metro Areas by Home Price Appreciation

	Metro	Price Change (%)	Pop. Change Rate (per 1,000)
Top 5	Boise City, ID	+58.2	+42.2
	Coeur d’Alene, ID	+56.0	+51.2
	Idaho Falls, ID	+55.5	+35.1
	Austin, TX	+50.6	+32.7
	Twin Falls, ID	+50.4	+24.6
Bottom 5	Houma, LA	+1.4	−2.8
	Shreveport, LA	+1.4	−11.0
	Odessa, TX	+0.6	−26.7
	Midland, TX	−1.2	−10.3
	Lake Charles, LA	−2.1	−54.9

Sources: Zillow ZHVI; Census Bureau PEP Vintage 2022.

Figure 1 shows the full cross-metro relationship. Sun Belt and Mountain West destinations cluster in the upper-right quadrant while major coastal urban cores sit in the lower-left. Figure 2 illustrates the top and bottom 15 metros with bar color encoding the population change rate, confirming that high-appreciation metros are systematically blue (in-migration) and low-appreciation metros are red or gray. Figure 3 shows monthly ZHVI trajectories: pandemic destinations accelerate sharply after March 2020 while major origin cities flatten or decline through mid-2021, with the divergence beginning almost immediately after lockdowns were declared.

4 Robustness

Table 3 reports the migration coefficient across five alternative specifications. Several concerns motivate these checks: (1) Idaho’s cluster of high-appreciation small markets may be amplifying the result; (2) Lake Charles, LA (Hurricane Laura displacement) and Odessa/Midland, TX (oil price collapse) represent non-pandemic shocks that contaminate the left tail; (3) heteroskedasticity may bias standard errors; and (4) thin small markets may drive the result more than economically significant metros.

Table 3: Robustness Checks: Migration Coefficient Across Specifications

Specification	Coefficient	Std. Error	R^2
(2): Preferred (full sample, $n = 380$)	0.357***	0.027	0.366
(2): Heteroskedasticity-robust SEs	0.357***	0.034	0.366
(2): Idaho Excluded ($n = 371$)	0.318***	0.026	0.324
(2): Hurricane/Oil outliers Excluded ($n = 377$)	0.340***	0.027	0.345
(2): Large metros only >250k ($n = 188$)	0.333***	0.032	0.394
<i>Partial R^2 decomposition (full sample):</i>			
Price level only			$R^2 = 0.064$
Adding migration (M2)			$R^2 = 0.366$
Unique contribution of migration			$\Delta R^2 = 0.302$

*** $p < 0.001$. All specifications include $\ln(\text{Pre-pandemic ZHVI})$ as control.

Sources: Zillow ZHVI; Census Bureau PEP Vintage 2022.

The migration coefficient ranges from 0.318 to 0.357 across all specifications (a spread of less than 0.04) and remains significant at $p < 10^{-20}$ in every case. Three findings are particularly notable.

First, heteroskedasticity-robust standard errors widen the SE from 0.027 to 0.034 (a 25% increase) but the coefficient remains highly significant, confirming that inference does not depend on the constant variance assumption. The Breusch-Pagan test ($p = 0.096$) also fails to reject homoskedasticity at conventional levels, suggesting that heteroskedasticity is not severe, or of great concern.

Second, restricting to large metros (population > 250,000) yields a coefficient of 0.333 with a higher R^2 of 0.394. If smaller Idaho (or other Mountain West) markets were spuriously inflating the result, dropping them should decrease the coefficient, instead it holds, and R^2 rises. The migration–price relationship is if anything *stronger* among economically significant metros, directly addressing the concern that thin small markets are driving the finding.

Third, the partial R^2 decomposition is striking: pre-pandemic price level alone explains only 6.4% of home price appreciation variance across metros. Adding migration raises this to 36.6%, meaning migration

contributes 30.2 percentage points of explanatory power independently of price level. This is the cleanest statement of migration’s central role in pandemic-era home price dynamics.

5 Limitations and Extensions

Identification: This analysis makes use of cross-sectional OLS, so the relationship is largely correlational, not causal. Both migration and price appreciation may be jointly driven by remote work adoption rates by occupation mix or local amenity demand. A cleaner identification strategy might instrument for migration using pre-pandemic remote-work feasibility scores interacted with local broadband availability.

Supply elasticity: Markets with inelastic housing supply face larger price responses to the same demand shock. Adding the housing supply elasticity index as a control would sharpen the migration coefficient and help separate demand-side from supply-side explanations.

Migration decomposition: NPOPCHG conflates domestic migration, international migration, and natural increase. Decomposing using ACS 1-year flow tables or IRS SOI county-to-county data would isolate the domestic migration signal cleanly.

Panel specification: A panel with annual data and metro fixed effects would capture the gradual unfolding of migration flows into 2022 and absorb time-invariant confounders that cross-sectional OLS cannot.

Small market dominance: The top 15 appreciating metros are disproportionately small Idaho and Mountain West markets. These results are real but partly reflect low base prices and thin markets where a modest number of movers produces large percentage gains. A robustness check using absolute dollar appreciation as the outcome should reduce the influence of these markets.

Figures/Appendix

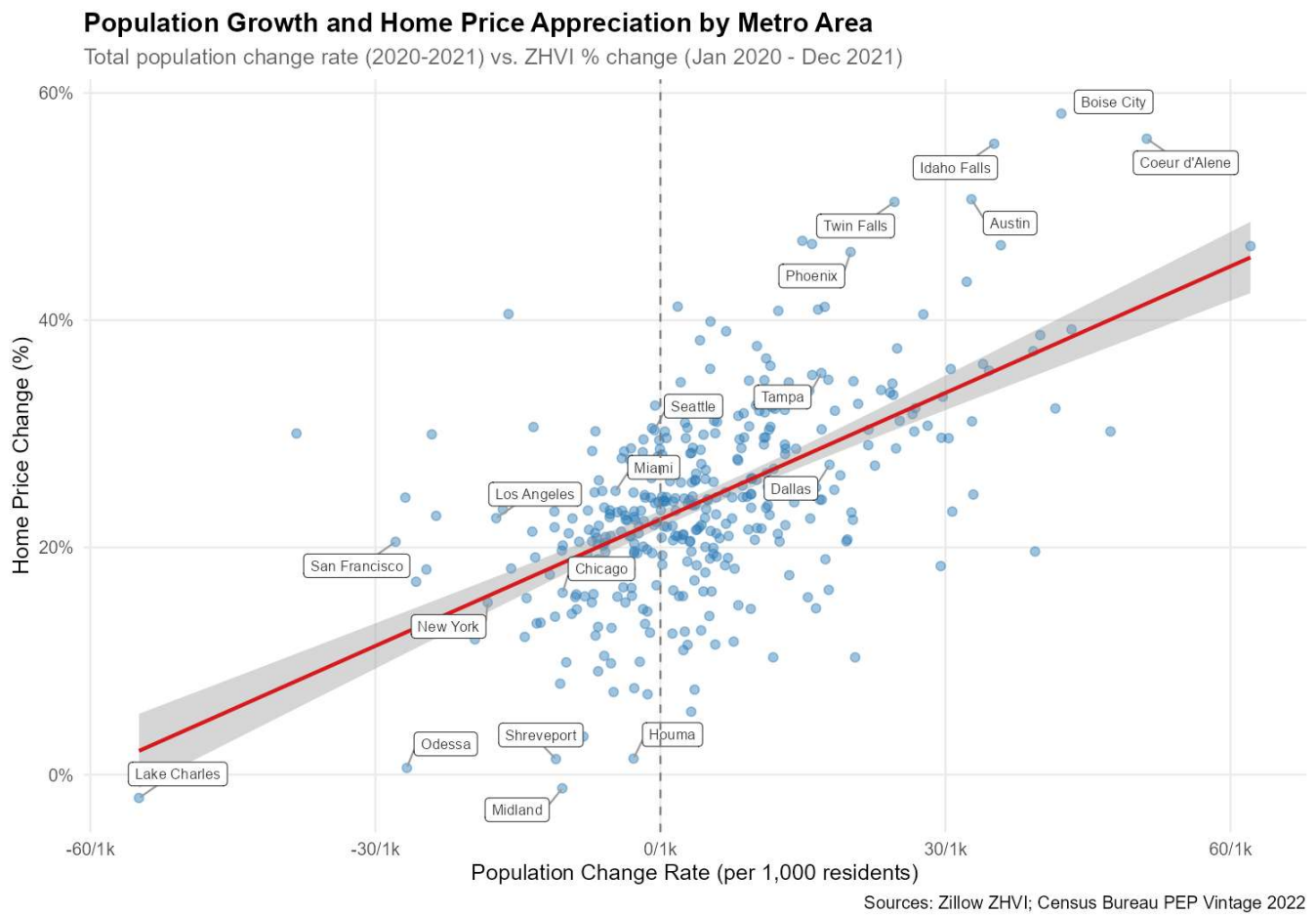


Figure 1: Population growth rate (per 1,000 residents, 2020–2021) vs. home price appreciation (Jan 2020 – Dec 2021) across 380 MSAs. Red line shows OLS fit with 95% confidence band. Correlation = 0.576. Sources: Zillow ZHVI; Census Bureau PEP Vintage 2022.

Home Price Appreciation: Top and Bottom 15 Metro Areas

Bar color = population change rate (blue = growth, red = decline)

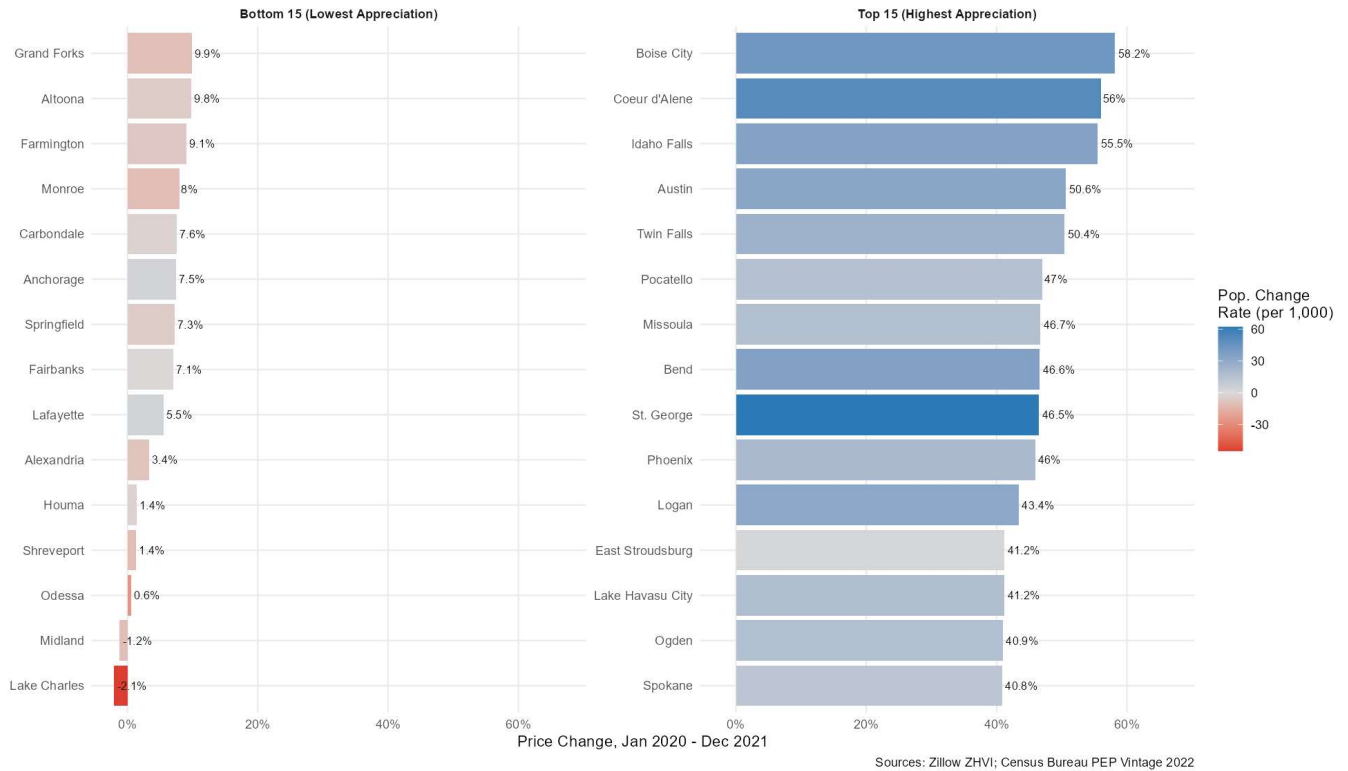


Figure 2: Home price appreciation for the top and bottom 15 metro areas. Bar color encodes population change rate: blue = population growth, red = population decline. Sources: Zillow ZHVI; Census Bureau PEP Vintage 2022.

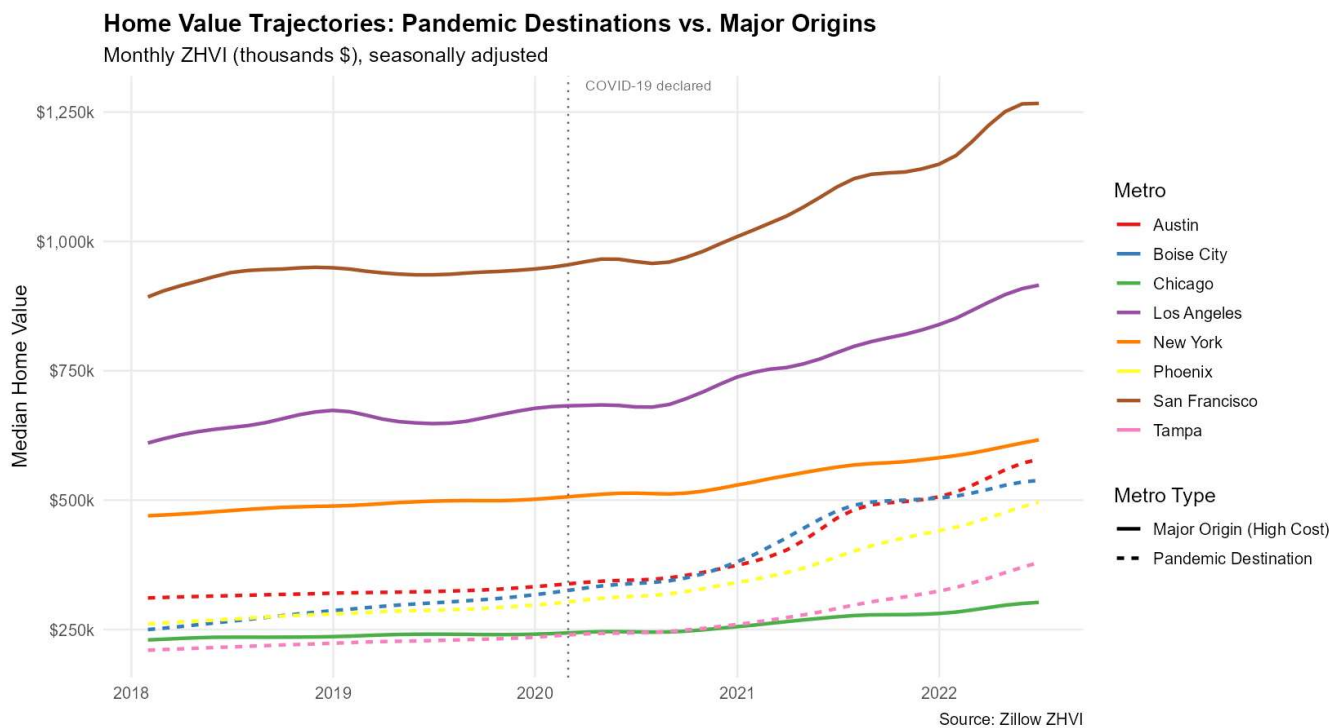


Figure 3: Monthly ZHVI trajectories (seasonally adjusted, thousands USD) for selected pandemic destinations (dashed lines) vs. major origin cities (solid lines). Vertical dotted line marks March 2020 (COVID-19 declared). Source: Zillow ZHVI.

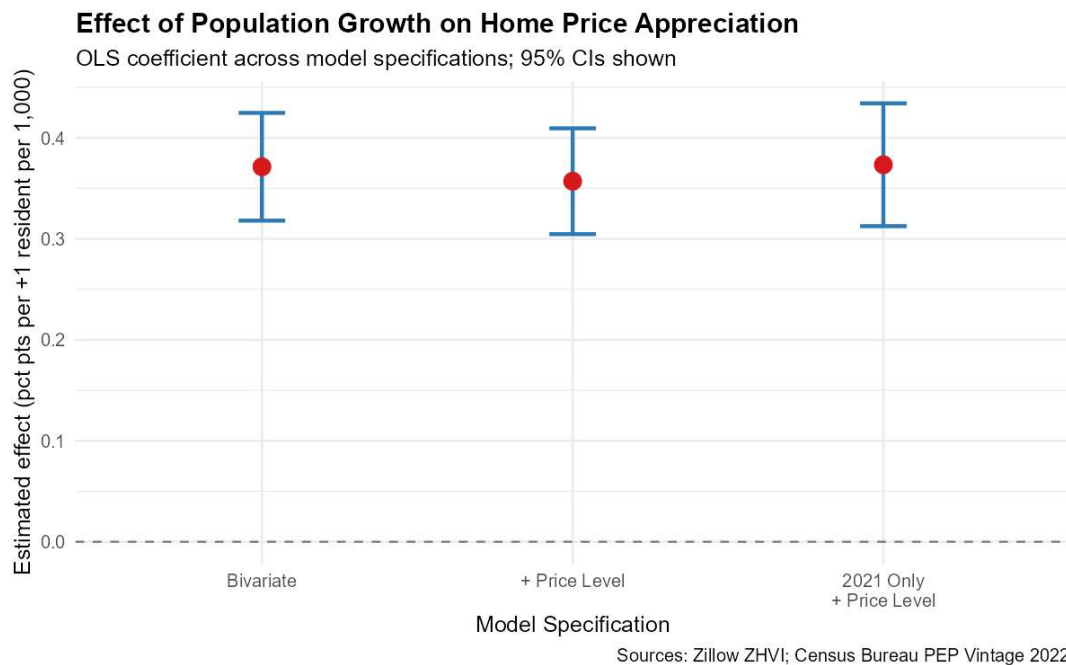


Figure 4: OLS coefficient on population change rate across three model specifications with 95% confidence intervals. The estimate is stable at ≈ 0.36 – 0.37 across all specifications. Sources: Zillow ZHVI; Census Bureau PEP Vintage 2022.

References

- Zillow Research. (2024). "Zillow Home Value Index (ZHVI): Metro-Level, All Homes, Middle Tier, Smoothed and Seasonally Adjusted." Zillow, January 16, 2026. <https://www.zillow.com/research/data/>.
- U.S. Census Bureau. (2023). "Population and Housing Unit Estimates: Vintage 2022 Census Bureau Metropolitan and Micropolitan Statistical Area Totals." U.S. Census Bureau. Accessed May 21, 2026. <https://www2.census.gov/programs-surveys/popest/datasets/2020-2022/metro/totals/cbsa-est2022.csv>.